

Ex. No. 6 — Inference of Q Factor of Rectangular Cavity Resonator for TE

101

Mode with the Following Inputs: a) Dimension a in x-axis b) Dimension b in y-axis c) Dimension c in z-axis

Test Case 1

Design a rectangular cavity resonator operating in the TE₁₀₁ mode with a = 5 cm, b = 4 cm, c = 10 cm, made of

101

copper ($\sigma = 5.8 \times 10^7$ S/m). Examine how the Q-factor changes as dimension c is varied from 8 cm to 12 cm, with a = 4 cm and b = 3 cm held constant.

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2; // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*pi*10^-7; f101 = 3.335*10^9; d =
sqrt(1/(%pi*f101*Uo*C)); Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 +
c^3)+a*c*(a*a+c*c))); disp(Q, 'Quality factor of TE101 = ');
```

Result Table

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
4.0	3.0	8.0	11155.6576	11155.6576
4.0	3.0	9.0	11096.4312	11096.4312
4.0	3.0	10.0	11034.2245	11034.2245
4.0	3.0	11.0	10975.7477	10975.7477
4.0	3.0	12.0	10923.2481	10923.2481

The screenshot displays the Scilab 2026.1.0 interface. The top toolbar contains various icons for file operations, editing, and execution. Below the toolbar, the title bar indicates the active window is 'exp 6.sci (C:\Users\salttransmission line\exp 6.sci) - SciNotes'. The main editor area shows a script with the following code:

```
1 clear;
2 clc;
3 a=5*10^-2,b=4*10^-2,c=10*10^-2,C=5.8*10^7,Uo=4*pi*10^-7;
4 f101=3.335*10^9;
5 d=sqrt(1/(%pi*f101*Uo*C));
6 Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
7 disp(Q,'Quality factor of TE101 = ');
```

On the right side, the 'Scilab 2026.1.0 Console' window is open, showing the output of the script:

```
14325.571
"Quality factor of TE101 = "
```

Below the console, a prompt '-->' is visible.

Test Case 2

Result: The Q-factor of the rectangular cavity resonator for the TE₁₀₁ mode, obtained by varying the dimensions, is determined using Scilab. Increasing c slightly decreases the Q-factor here, since a larger c raises the wall surface area faster than it raises the stored volume for this a, b combination.

Optimize the rectangular cavity resonator to achieve the highest possible Q-factor, based on the ranges of a, b and c explored in Test Cases 1, 3 and 4 (a: 4–6 cm, b: 3–5 cm, c: 8–12 cm).

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2; // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*pi*10^-7; f101 = 3.335*10^9; d =
sqrt(1/(pi*f101*Uo*C)); Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 +
c^3)+a*c*(a*a+c*c))); disp(Q, 'Quality factor of TE101 = ');
```

Result Table — Highest Q-Factor Combination

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
6.0	5.0	8.0	17361.4539	17361.4539

The maximum a (6 cm), maximum b (5 cm) and minimum c (8 cm) within the explored ranges give the highest Q-factor: Q rises with a and b (more stored volume relative to the wall area facing those axes) and falls as c grows over this range, so the best combination sits at the a/b upper bound and the c lower bound.

exp 6.2.sci (C:\Users\sal\transmission line\exp 6.2.sci) - SciNotes

exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X exp 3.2.sci X exp 4.2.sci X exp 5.2.sci X

exp 1.1.sci X exp 2.sci X exp 3.sci X exp 4.sci X

exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X exp 6.3.sci X exp 6.4.sci X exp 1.2.sci X

```
1 clear;
2 clc;
3 a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*pi*10^-7;
4 f101=3.335*10^9;
5 d=sqrt(1/(pi*f101*Uo*C));
6 Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
7 disp(Q, 'Quality factor of TE101 = ');
8
```

Scilab 2026.1.0 Console

14325.571

"Quality factor of TE101 = "

-->

Test Case 3

Assess how a single dimension affects resonator performance: fix $b = 4$ cm and $c = 10$ cm, and vary a between 4 cm and 6 cm in equal steps.

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2; // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*pi*10^-7; f101 = 3.335*10^9; d = sqrt(1/(pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c))); disp(Q, 'Quality factor
of TE101 = '); Q-Factor for Varying a
```

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
4.0	4.0	10.0	12331.8421	12331.8421
4.5	4.0	10.0	13376.4936	13376.4936
5.0	4.0	10.0	14325.5713	14325.5713
5.5	4.0	10.0	15180.8629	15180.8629
6.0	4.0	10.0	15945.2065	15945.2065

The Q-factor increases steadily as a increases from 4 cm to 6 cm, showing the resonator is fairly sensitive to a . In practice, the largest dimension generally has the greatest influence on Q , since it dominates the volume-to-surface-area ratio that Q depends on.

exp 6.2.sci (C:\Users\salttransmission line\exp 6.2.sci) - SciNotes

exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X exp 3.2.sci X exp 4.2.sci X exp 5.2.sci X

exp 1.1.sci X exp 2.sci X exp 3.sci X exp 4.sci X

exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X exp 6.3.sci X exp 6.4.sci X exp 1.2.sci X

```
1 clear;
2 clc;
3 a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*pi*10^-7;
4 f101=3.335*10^9;
5 d=sqrt(1/(pi*f101*Uo*C));
6 Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
7 disp(Q, 'Quality factor of TE101 = ');
8
```

Scilab 2026.1.0 Console

14325.571

"Quality factor of TE101 = "

-->

Test Case 4

Investigate how dimension b affects the Q-factor: fix a = 5 cm and c = 10 cm, and vary b between 3 cm and 5 cm.

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2; // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*pi*10^-7; f101 = 3.335*10^9; d =
sqrt(1/(pi*f101*Uo*C)); Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 +
c^3)+a*c*(a*a+c*c))); disp(Q, 'Quality factor of TE101 = ');
```

Q-Factor for Varying b

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
5.0	3.0	10.0	12603.7478	12603.7478
5.0	3.5	10.0	13533.2277	13533.2277
5.0	4.0	10.0	14325.5713	14325.5713
5.0	4.5	10.0	15009.0432	15009.0432
5.0	5.0	10.0	15604.6401	15604.6401

Result The Q-factor of a rectangular cavity resonator operating in the TE mode was determined using Scilab for a 101 range of dimensions a, b and c. Q increases with both a and b and decreases with c over the ranges studied, and the highest Q-factor within the explored ranges was obtained at a = 6 cm, b = 5 cm, c = 8 cm.

The screenshot displays the Scilab 2026.1.0 environment. The main window shows a script editor with the following code:

```
1 clear;
2 clc;
3 a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*pi*10^-7;
4 f101=3.335*10^9;
5 d=sqrt(1/(pi*f101*Uo*C));
6 Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
7 disp(Q, 'Quality factor of TE101 = ');
8
```

The console window on the right shows the output of the script:

```
14325.571
"Quality factor of TE101 = "
```

Test Case 5